

INNOTRACK INTERNSHIP PROGRAM - 2026



- **Name – S. Shalini**
- **Project Name – Real Time Heart Rate Anomaly**

Detector

- **Theme – Health Care**
- **VTU ID – 27976**



Real Time Heart Rate Anomaly Detector



INTRODUCTION:

Heart rate monitoring plays a crucial role in maintaining cardiovascular health and detecting potential medical conditions. Variations in heart rate patterns may indicate abnormalities such as tachycardia, bradycardia, arrhythmia, or other health concerns. Early detection of heart rate anomalies can assist healthcare professionals in providing timely medical intervention and improving patient outcomes. The Real Time Heart Rate Anomaly Detector is a Machine Learning based web application that analyzes heart rate data and identifies abnormal patterns in real time. The system combines predictive analytics and a user-friendly web interface to provide continuous heart rate monitoring and support data-driven healthcare decision-making.

Idea :

- Develop a machine learning-based system to detect heart rate anomalies in real time.
- Support healthcare monitoring and early diagnosis through data-driven insights..

Solution :

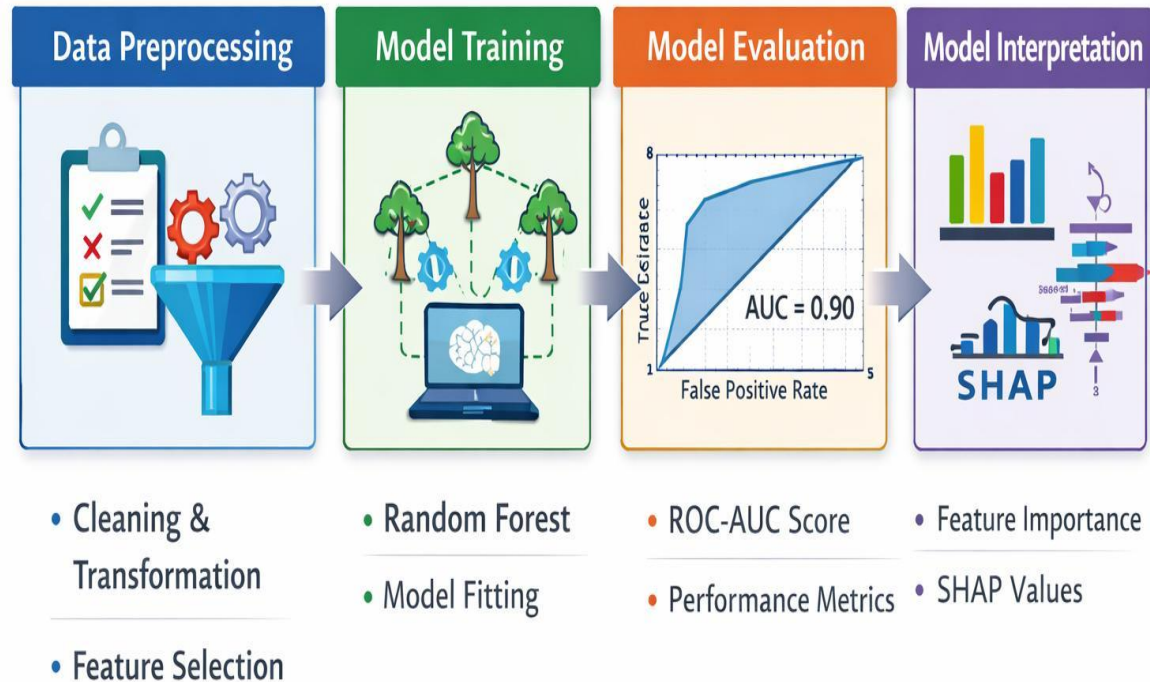
- Built a Gradient Boosting model to classify heart rate patterns as Normal or Abnormal.
- Developed a Flask web application with real-time monitoring, history tracking, and analytics features.

Uniqueness:

- Real-time heart rate anomaly detection.
- Interactive analytics dashboard.
- Monitoring history tracking and filtering.
- User authentication and profile management.
- Data-driven decision support for healthcare professionals.
- Centralized platform for heart rate analysis and anomaly assessment.
- Secure storage of monitoring records using SQLite.
- Confidence score generation for each prediction.

TECHNICAL APPROACH

Machine Learning Pipeline for Healthcare Data



Technologies to be Used:

- **Frontend** – HTML, CSS3, JavaScript
- **Backend** – Python, Flask Framework
- **Database** – SQLite
- **Tools** – Visual Studio Code , GitHub
- **ML** - Gradient Boosting Classifier

Methodology:

- Processed heart rate data and trained a Gradient Boosting model to detect abnormal heart rate patterns..
- Integrated the trained model into a Flask web application for real-time monitoring, history tracking, and analytics..

FEASIBILITY AND VIABILITY

- Developed using open-source tools and technologies.
- Requires minimal hardware and software resources.
- Low development, deployment, and maintenance cost.
- Easy to integrate with existing healthcare monitoring systems.
- Supports real-time heart rate anomaly detection.
- Suitable for healthcare and smart health applications.
- Provides practical value for healthcare professionals and analysts.
- Creates opportunities for future expansion into wearable and IoT-based applications
- Reduces manual effort in heart rate monitoring and anomaly assessment.

CHALLENGES FACED

- Handling missing and inconsistent values in the heart rate dataset.
- Managing variations between normal and abnormal heart rate patterns.
- Integrating the trained machine learning model with the Flask web application.
- Designing a professional and responsive user interface.
- Implementing user authentication and profile management features.
- Ensuring accurate real-time detection and database storage
- Creating meaningful analytics and visualization dashboards.
- Maintaining consistency between training, testing, and deployment environments..

FUTURE ENHANCEMENTS

- Integrating real-time heart rate data from wearable health devices.
- Enhancing detection accuracy using advanced machine learning and deep learning models.
- Integrating additional health parameters for comprehensive monitoring.
- Developing a mobile application for health monitoring and alerts.
- Implementing personalized health recommendations and notifications.
- Adding cloud-based health record management features.
- Creating advanced analytics and visualization dashboards.
- Deploying the system for large-scale healthcare monitoring applications.

IMPACT AND BENEFITS

- Enables early detection of heart rate anomalies using machine learning.
- Supports healthcare professionals in making informed decisions.
- Provides instant anomaly detection for effective health monitoring.
- Helps identify heart rate patterns and health trends from historical data.
- Improves healthcare planning and resource allocation.
- Supports smart healthcare development through data-driven solutions.
- Provides an easy-to-use platform for heart rate analysis and reporting.
- Demonstrates the practical application of AI and Machine Learning in healthcare.
- Enhances awareness of factors affecting cardiovascular health.

CONCLUSION

The Real Time Heart Rate Anomaly Detector successfully demonstrates the integration of Machine Learning and Web Development to address a real-world healthcare challenge. The system effectively detects heart rate anomalies based on heart rate patterns and physiological data, providing quick and reliable insights for better decision-making. This project highlights the importance of data-driven solutions in healthcare monitoring and supports the vision of smart healthcare systems. By combining predictive analytics, a user-friendly interface, and database integration, the system offers a complete end-to-end solution that can be further enhanced for real-time deployment and large-scale healthcare applications in the future.

THANK YOU